Conference Management System - Relational Schema

Relational Schema

Users (user_id PK, full_name, email UNIQUE, age, role, created_at)

Credentials (user_id PK, FK → Users.user_id, password_hash)

Conferences (conference_id PK, title, description, location, start_time, end_time, capacity, created_at)

Registrations (registration_id PK, user_id FK → Users.user_id, conference_id FK → Conferences.conference_id, status, registered_at)

WaitingList (waiting_id PK, registration_id FK → Registrations.registration_id, UNIQUE, joined_at)

Database Constraints

1. Primary Keys (PK)

Each table has a primary key that uniquely identifies its records.

2. Foreign Keys (FK)

- Credentials references Users.
- Registrations references Users and Conferences.
- WaitingList references Registrations.

3. Unique Constraints

- Users.email must be unique.
- Registrations(user_id, conference_id) must be unique.
- WaitingList.registration_id must be unique.

4. Check Constraints

- Users.age > 0
- Users.role must be ADMIN or PARTICIPANT.
- Conferences.capacity > 0
- Conferences.end_time must be later than start_time.
- Registrations.status must be CONFIRMED, WAITLISTED, or CANCELLED.

5. Required Fields

Fields marked NOT NULL in schema.sql cannot contain NULL values.

6. Cascading Deletes

- Deleting a user also deletes their credentials.
- Deleting a registration also deletes its waiting-list entry.